Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Configuring VMware Carbon Black Cloud Connectors

This connector allows Stellar Cyber to ingest VMware Carbon Black Cloud alerts from an AWS S3 bucket and add the alerts to the data lake.

It also supports the Contain Host response action.

Stellar Cyber connectors with the Collect function (collectors) may skip collecting some data when the ingestion volume is large, which potentially can lead to data loss. This can happen when the processing capacity of the collector is exceeded.

Connector Overview: VMware Carbon Black Cloud

Capabilities

- Collect: **Yes**
- Respond: **Yes**
- Native Alerts Mapped: **Yes**
- Runs on: **DP**
- Interval: **Configurable**

Collected Data

Content Type	Index	Locating Records

Alert
Endpoint Event 6.5.0
	Syslog
	Assets (for Alert)
		msg_class:
		`carbonblack_alert`
		`carbonblack_endpoint_event` 6.5.0
		
		msg_origin.source:
		`carbonblack_cloud`

Content Type	Index	Locating Records
		msg_origin.vendor: `vmware_carbonblack` msg_origin.category: `endpoint`

If there is asset information in the logs for Events and Alerts, the detected assets are reported in the Assets Index.

Domain

N/A

Response Actions

Action	Required Fields
Contain Host	`device_id`

The required permissions for response actions are `device.quarantine` (Execute), `device` (Read), and `device.bg-scan` (Execute).

Third Party Native Alert Integration Details

Alerts of type CB Analytics, Device Control, and Watchlist. Select the Alert content type.

For details, see Integration of Third Party Native Alerts.

Required Credentials and Configurations

- For Collect: Carbon Black Organization Key, Bucket, Access Key ID, Secret Access Key, Alert Prefix for Alert Collection, and Endpoint Event Prefix for Endpoint Event Collection
- For Respond: Carbon Black Organization Key, Host, API secret key, and API ID

Locating Records

To search the alerts in the Alerts index or to search the Original Records in the Syslog index, use the query: `msg_origin.source:carbonblack_cloud AND (event.type:(NON_MALWARE OR NEW_MALWARE OR KNOWN_MALWARE OR RISKY_PROGRAM) OR (NOT _exists_:event.type AND vmware_carbonblack.type: (CB_ANALYTICS OR DEVICE_CONTROL)))`

Adding a VMware Carbon Black Cloud Connector

To add a VMware Carbon Black Cloud connector:

1. Obtain the VMware Carbon Black organization ID
2. Gather / Configure AWS S3 information
3. (Optional) Configure Carbon Black for Respond action
4. Add the connector in Stellar Cyber
5. Test the connector
6. Verify ingestion

Obtaining the VMware Carbon Black Organization ID

Use of this connector requires that you have previously configured your VMware Carbon Black deployment to send data to AWS S3. You still require the **Organization ID** to configure the connector. You can find this in your VMware Carbon Black Cloud portal under **Settings | API Access | API Keys**.

Gathering / Configuring the AWS S3 Information

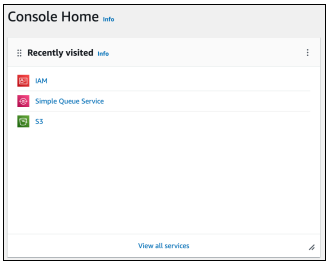
Use this section to obtain the following required details for use in configuring your connector:

- **Access Key ID**
- **Secret Access Key**
- **Bucket**
- **Alert Prefix**
- **Endpoint Event Prefix**

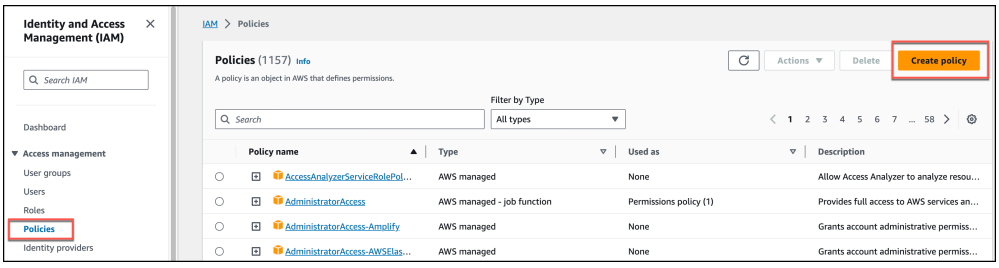
Creating a User

To add a user with the appropriate permissions:

1. Log in to your **AWS Management Console** at https://aws.amazon.com/console. View the services in the Console Home or choose **View all services**.



2. Choose **IAM**. The **IAM Dashboard** appears.
3. Choose **Policies** and then choose **Create Policy**.



4. In the **Create policy** pane, choose the **JSON** tab.



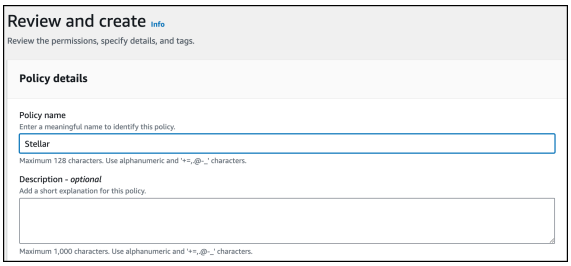
5. Using the example below as a guide, edit the JSON policy document.

The following is just an EXAMPLE. You must modify this JSON to match the resources in your environment.

VMware Carbon Black policy

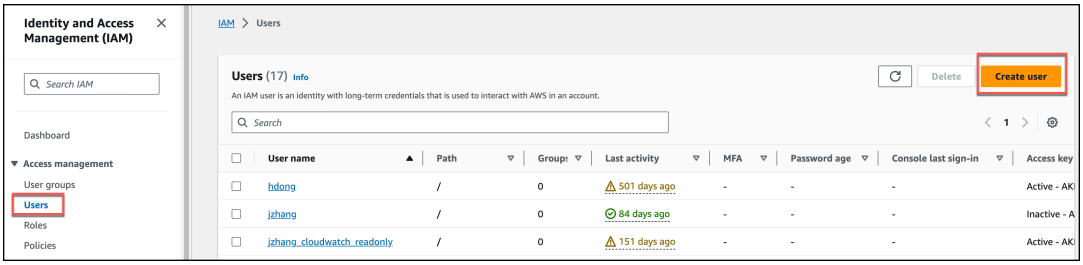
```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "s3:GetObject",
                "s3:ListBucket"
            ],
            "Resource": [
                "arn:aws:s3:::stellarcarbonblack/*",
                "arn:aws:s3:::stellarcarbonblack"
            ]
        }
    ]
}
```

6. Choose **Next**.
7. Give your policy a name to associate it with Stellar Cyber, then choose **Create policy**.

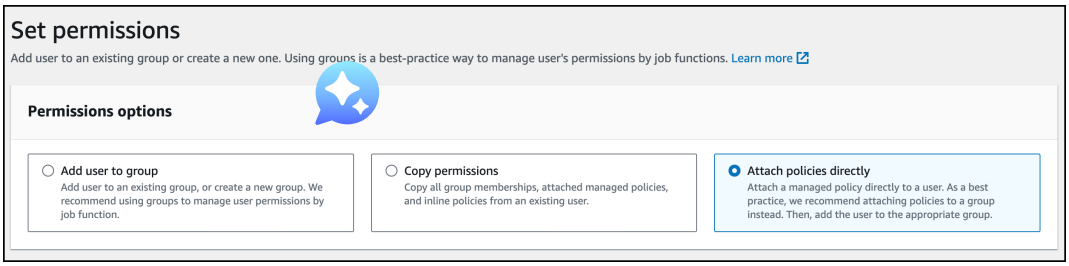


The policy can now be attached to a user.

8. From the IAM navigation pane, choose **Users** and then choose **Create user**.



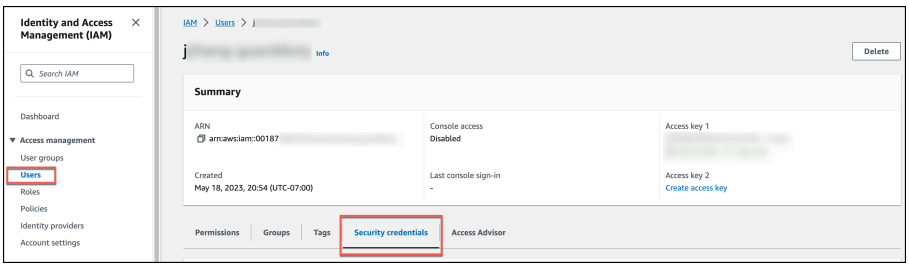
9. On the **Specify user details** page for **User details**, enter a **User name** for the new user.
10. Choose **Provide user access to the – AWS Management Console** *optional* to produce login credentials for the new user, such as a password.
11. Choose how you want to create the Console password and then choose **Next**.
12. On the **Set permissions** page, choose **Attach policies directly**. Then search for the policy you created above.



13. Select the checkbox to the left of the policy name, then choose **Next**.
14. On the **Review and create** page, verify the information and then choose **Create user**.

Creating an Access Key

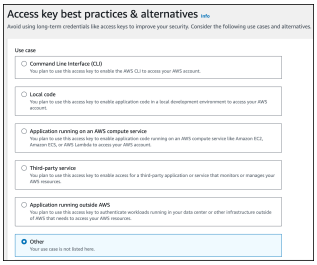
1. Log in to the IAM Console with your AWS account ID, your IAM user name, and password. You must have the permissions required to create access keys for a user.
2. Choose **Users**, then click the user name, and click **Security credentials**.



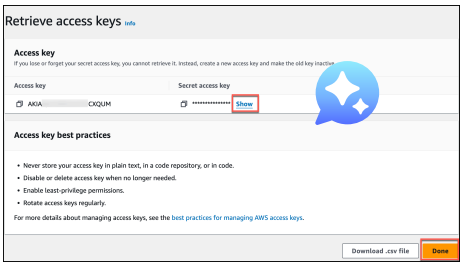
3. In the **Access keys** section, choose **Create access key**. Access keys have two parts: an access key and a secret access key. Click **Create access key**.



4. On the **Access key best practices & alternatives** page, choose **Other** and then choose **Next**.



5. (Optional) On the **Set description tag** page, enter a description.
6. Click **Create access key**.
7. On the **Retrieve access keys** page, choose **Show** to reveal the value of the user's secret access key. Save the access key and secret access key in a secure location. You will need them when configuring the connector in Stellar Cyber.



8. Click **Done**.

Creating and Configuring the S3 Bucket

1. Navigate to the **AWS S3 Console** (https://s3.console.aws.amazon.com/s3/home).
2. From the navigation pane, select **Bucket** to ensure the buckets dashboard is displayed.
3. Click **Create bucket**. Give the bucket a unique name that does not contain uppercase letters or underscores; dashes are allowed.
4. Verify the **AWS region** is set to the desired value (for example, US East (N. Virginia) us-east-1).
5. Ensure the check-box for **Block all Public Access** is enabled/checked. The connector does not require public access set.
6. At the bottom of the screen, click the button to **Create bucket**.
7. From the AWS S3 list of buckets, click the name of the bucket you just created.
8. From the bucket details pane, click the button to **Create folder**. This is the folder to which you will send data from the **VMware Carbon Black console**. Note the precise folder name. You will use this folder name to replace the prefix in the bucket policy in the next step, in the connector, and when you add a Data Forwarder in the **Carbon Black Cloud console**.

   Each VMware Carbon Black Data Forwarder requires its own folder. Otherwise, data from multiple forwarders can mix in the same folder and impair parsing of data.

9. Now, click the **Permissions** tab and click the button to **Edit** the **Bucket Policy**.
10. Modify the **Resource** parameter as indicated below to include your Carbon Black product region and the name of the folder you created above. This pane includes buttons to sample policies and to an AWS Policy Generator tool you can use to create the JSON for your bucket policy.

```
{
        "Version": "2012-10-17",
        "Statement": [
                {
                        "Sid": "Statement1",
                        "Principal": {},
                        "Effect": "Allow",
                        "Action": [],
                        "Resource": []
                }
```

```
        ]
}
```

Following is an example of a complete policy with the region, folder, and allowed actions specified, using the AWS Policy Generator tool. The most critical element to include is the Resource field.

```
{
  "Version": "2012-10-17",
  "Id": "Policy1647298852280",
  "Statement": [
    {
      "Sid": "Stmt1647298832372",
      "Effect": "Allow",
      "Principal": {
        "AWS":
          "arn:aws:iam::132308400445:role/mcs-psc-prod-event-forwarder-us-
east-1-event-forwarder"
        },
      "Action": [
        "s3:PutObject",
        "s3:PutObjectAcl"
      ],
      "Resource":
      "arn:aws:s3:::bucket-name/prefix-folder-name/*"
    }
  ]
}
```

11. Repeat Step 8 to 10 for each data type you want to collect: once for alerts and once for endpoint events, ensuring they are in the same bucket.

At this point you should have made note of the following information from your VMware and AWS steps, above.

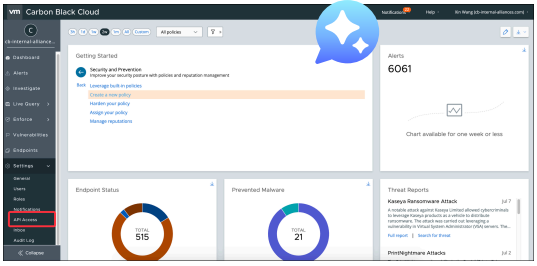
- **Organization ID**
- **Access Key ID**
- **Secret Access Key**
- **Bucket**
- **Alert Prefix**
- **Endpoint Event Prefix**

(Optional) Configuring Carbon Black for Respond Action

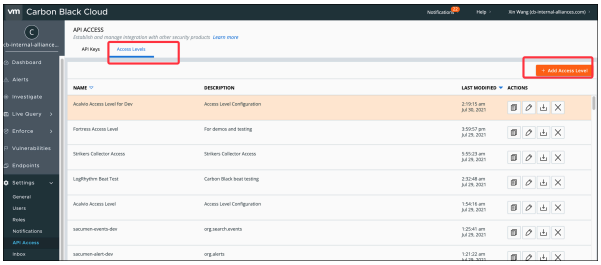
To configure settings in VMware Carbon Black for the Contain Host respond action:

1. Log in to VMware Carbon Black.

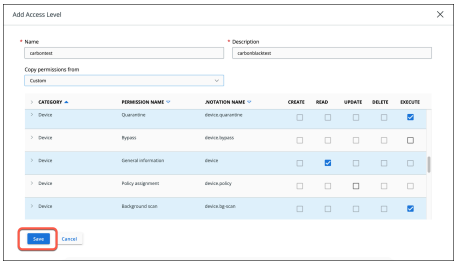
2. Expand **Settings**, then click **API Access**.



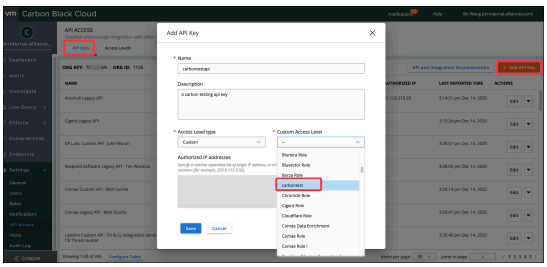
3. Click **Access Levels**.



4. Click **Add Access Level**.

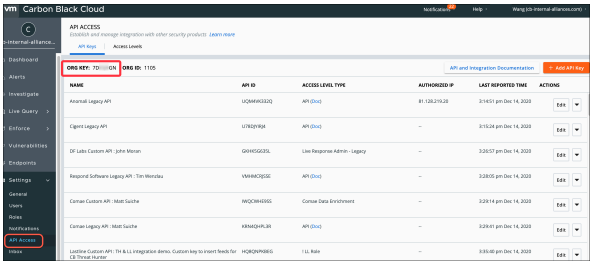


5. Enter a **Name** and **Description**.
6. Select permissions as follows to allow the API key you create to have permissions for the response action:
    - For Device > Quarantine (device.quarantine), select **Execute**.
    - For Device > General information (device), select **Read**.
    - For Device > Background scan (device.bg-scan), select **Execute**.
7. Click **Save**.
8. After you create the **Access Level**, click **API Keys** and click **Add API Key**.



9. Enter a **Name** and **Description**.
10. For **Access Level type**, select **Custom**.
11. Select the name of the **Custom Access Level** you previously created.
12. (Optional) If you want to restrict the API key to only work on certain IP addresses, you can enter them in **Authorized IP addresses**.

13. Click **Save**.
14. Copy your **API ID** and **Secret Key**. You will use them when configuring the connector in Stellar Cyber.
15. Locate your **ORG KEY** as follows:

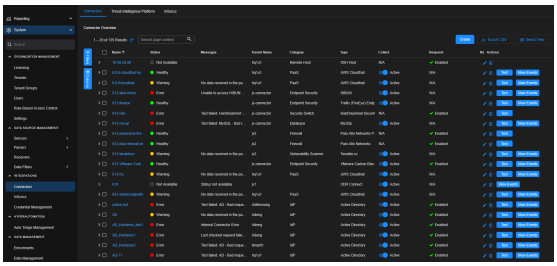


16. Make note of the **ORG KEY**. You will use it when configuring the connector in Stellar Cyber.

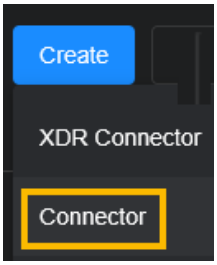
Adding the Connector in Stellar Cyber

With the access information handy, you can add a VMware Carbon Black Cloud connector in Stellar Cyber:

1. Log in to Stellar Cyber.
2. Click **System** | **INTEGRATIONS** | **Connectors**. The **Connector Overview** appears.

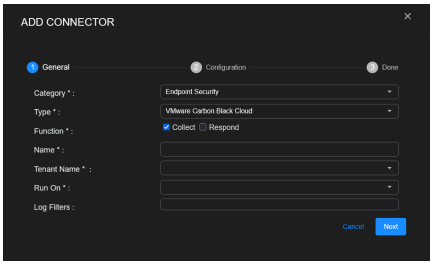


3. Click **Create** and select **Connector**.



The **General** tab of the **Add Connector** screen appears. The information on this tab cannot be changed after you add the connector.

4. Choose **Endpoint Security** from the **Category** dropdown.
5. Choose **VMware Carbon Black Cloud** from the **Type** dropdown.

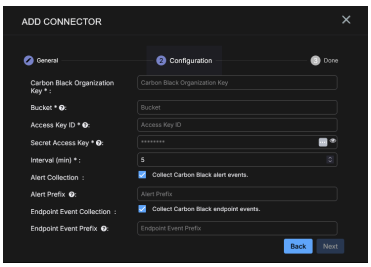


The asterisk (*) indicates a required field.

6. Choose the **Function**: **Collect** to collect logs; **Respond** for the respond action to contain hosts. You can select either **Collect** or **Respond** or both **Collect** and **Respond**. After you click **Next**, the configuration will vary, depending on your selection.
7. Enter a **Name**.

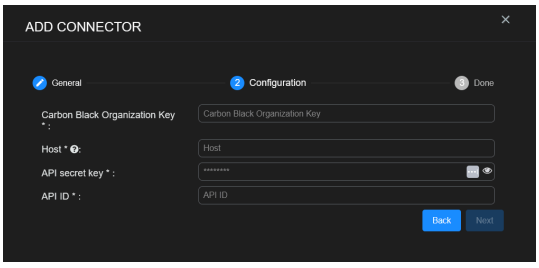
   Notes:
   - This field does not accept multibyte characters.
   - It is recommended that you follow a naming convention such as tenantname-connectortype.

8. Choose a **Tenant Name**. The Interflow records created by this connector include this tenant name.
9. Choose the device on which to run the connector.
10. (Optional) When the **Function** is **Collect**, you can apply **Log Filters**. For information, see Managing Log Filters.
11. Click **Next**. The following **Configuration** tab appears for the **Collect** function.



The asterisk (*) indicates a required field.

The following **Configuration** tab appears for the **Respond** function.



The asterisk (*) indicates a required field.

If you select both **Collect** and **Respond**, the **Configuration** tab contains the fields for both.

12. For the **Collect** function:
    a. Enter the **Carbon Black Cloud Organization Key** you noted above in Gathering / Configuring the AWS S3 Information.
    b. Enter the **Bucket**. This is the AWS S3 bucket you noted above, configured in Carbon Black Event Forwarder.
    c. Enter the **Access Key ID** you noted above.

d. Enter the **Secret Access Key** you noted above.

e. Choose the **Interval (min)**. This is how often the logs are collected.

f. For the logs for **Alert Collection**, select **Collect Carbon Black alert events**. 6.5.0

g. Enter the **Alert Prefix** you noted above, configured in Carbon Black Event Forwarder for alerts. (For this connector, it is the folder you created in the bucket above.)

You can just enter **alerts** if the S3 bucket has the following prefix syntax: `alerts/org_key=my_org/year=2022/month=11/day=23/hour=11/minute=56/`

h. For the logs for **Endpoint Event Collection**, select **Collect Carbon Black endpoint events**. 6.5.0

i. Enter the **Endpoint Event Prefix**, configured in Carbon Black Event Forwarder for endpoint events.

13. For the **Respond** function:

a. Enter the **Organization Key**. This is the **ORG KEY** you noted above in (Optional) Configuring Carbon Black for Respond Action.

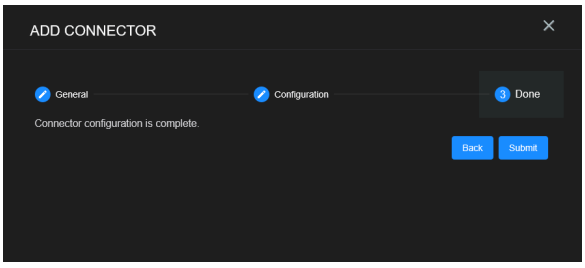
b. Enter your **Host**.

Do not include the https prefix.

c. Enter the **API secret key** you noted above.

d. Enter the **API ID** you noted above.

14. Click **Next**. The final confirmation tab appears.



15. Click **Submit**.

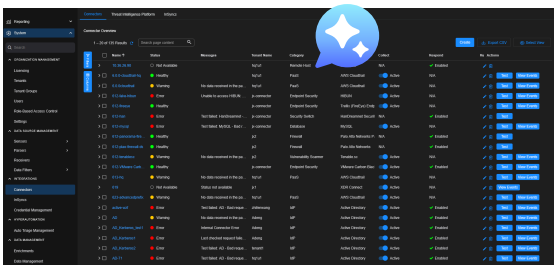
The new connector is immediately active and collects logs beginning today.

Testing the Connector

When you add (or edit) a connector, we recommend that you run a test to validate the connectivity parameters you entered. The test validates authentication and connectivity.

For connectors running on a sensor, Stellar Cyber recommends that you allow 30-60 seconds for new or modified configuration details to be propagated to the sensor before performing a test.

1. Click **System | INTEGRATIONS | Connectors**. The **Connector Overview** appears.



2. Locate the connector by name that you added, or modified, or that you want to test.
3. Click **Test** at the right side of that row. The test runs immediately.

Note that you may run only one test at a time.

Stellar Cyber conducts a basic connectivity test for the connector and reports a success or failure result. A successful test indicates that you entered all of the connector information correctly.

To aid troubleshooting your connector, the dialog remains open until you explicitly close it by using the **X** button. If the test fails, you can select the button from the same row to review and correct issues.

The connector status is updated every five (5) minutes. A successful test clears the connector status, but if issues persist, the status reverts to failed after a minute.

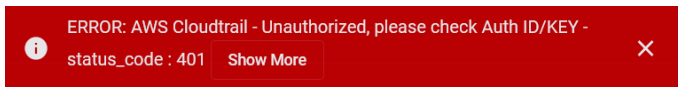
Repeat the test as needed.

Display sample messages...

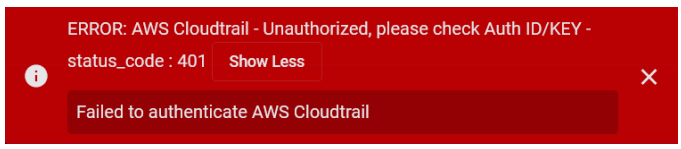
Success !



Failure with summary of issue:



Show More example detail:



If the test fails, the common HTTP status error codes are as follows:

HTTP Error Code	HTTP Standard Error Name	Explanation	Recommendation
400	Bad Request	This error occurs when there is an error in the connector configuration.	Did you configure the connector correctly?
401	Unauthorized	This error occurs when an authentication credential is invalid or when a user does not have sufficient privileges to access a specific API.	Did you enter your credentials correctly? Are your credentials expired? Are your credentials entitled or licensed for that specific resource?

HTTP Error Code	HTTP Standard Error Name	Explanation	Recommendation
403	Forbidden	This error occurs when the permission or scope is not correct in a valid credential.	Did you enter your credentials correctly? Do you have the required role or permissions for that credential?
404	Not Found	This error occurs when a URL path does not resolve to an entity.	Did you enter your API URL correctly?
429	Too Many Requests	This error occurs when the API server receives too much traffic or if a user's license or entitlement quota is exceeded.	The server or user license/quota will eventually recover. The connector will periodically retry the query. If this occurs unexpectedly or too often, work with your API provider to investigate the server limits, user licensing, or quotas.

For a full list of codes, refer to HTTP response status codes.

Verifying Ingestion

To verify ingestion:

1. Select **Threat Hunting**.

   The **Interflow Search** tab appears.

2. Change the **Indices** for the type of content you collected:

   - For **Alert** and **Endpoint Event**, change the **Indices** to **Syslog**.
   - For **Alert** only, change the **Indices** to **Assets**.

   The table immediately updates to show ingested Interflow records.

If you configured the connector Respond action, refer to **External Actions: Contain Host** to understand how to work with the Contain Host feature.

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